

StreamFlix — Content Analytics Project

Project Brief & Requirements — Data Analytics Coursework

Level: Intermediate · **Suggested time:** 3–4 weeks · **Tools:** Python / SQL / Power BI or Excel

1. Project Overview

Welcome to the StreamFlix Content Analytics project. You have been hired as a **Junior Data Analyst** at StreamFlix, a fast-growing subscription streaming service. The company's leadership team needs data-driven insights to decide what to watch, what to fund, and how to keep subscribers happy — and that is where you come in.

You will work with an industry-style dataset covering more than nine years of platform activity, containing roughly **979,000 records across 6 interconnected tables**. Your job is to clean, explore, analyse, and present the data so that non-technical managers can clearly understand what is happening on the platform.

1.1 Business Context

StreamFlix is a subscription video-on-demand (SVOD) platform serving subscribers across 20 countries. Its catalogue spans 16 genres — Drama, Comedy, Action, Thriller, Documentary, Animation, Horror, Romance, Sci-Fi, Fantasy, Crime, Kids & Family, Musical, Stand-Up, Reality and Mystery — across both movies and multi-season series. Subscribers watch on six device types (Smart TV, Mobile, Laptop, Tablet, Game Console, Streaming Stick) and pay on one of three plans: **Basic with Ads (\$6.99)**, **Standard (\$15.49)** and **Premium (\$22.99)**.

Leadership has the following pressing questions they need answered this quarter:

- Which genres and titles are driving the most watch time, and which are underperforming?
- Are we losing too many subscribers to churn, and what does an at-risk subscriber look like?
- Which subscriber segments — by plan, age, and region — are the most valuable and most engaged?
- Are there seasonal viewing patterns we should plan our content releases around?
- Which countries and languages are our biggest and fastest-growing markets?
- Is our content investment efficient, or are some genres and titles soaking up budget without the watch time to justify it?

2. Dataset Description

You are provided **6 CSV files and 1 SQL schema file**. All files together form the StreamFlix database. Together they contain approximately **979,000 records**, with viewing history spanning 2017 to June 2026 and subscriber signups going back to 2016.

2.1 Files Provided

File	Records	Size	Description
subscribers.csv	15,000	1.5 MB	Subscriber profiles, plan, location and churn status

File	Records	Size	Description
titles.csv	9,000	1.9 MB	Content catalogue with genre, cast, licensing and popularity
watch_history.csv	650,000	50.7 MB	Every viewing session — the central fact table
ratings.csv	130,000	5.1 MB	Star ratings (1–5) given by subscribers to titles
reviews.csv	110,000	9.9 MB	Written reviews with sentiment and helpful votes
watchlist.csv	65,000	2.8 MB	Titles subscribers saved, and whether they later watched
schema_and_sql.sql	—	8 KB	SQL CREATE TABLE statements + 12 analytical queries

2.2 Key Columns Reference

Below are the most important columns you will use throughout this project. Refer to the full Data Dictionary for every column definition.

watch_history.csv — The Central Table

Column	Type	Example	Description
watch_id	BIGINT	1	Unique identifier for each viewing session
subscriber_id	VARCHAR	SUB107630	Links to subscribers.csv
title_id	VARCHAR	TTL205207	Links to titles.csv
watch_date	DATE	2024-01-01	Date the session took place
device	VARCHAR	Smart TV	Smart TV / Mobile / Laptop / Tablet / etc.
watch_duration_min	DECIMAL	826.7	Minutes actually watched in the session
completion_pct	DECIMAL	93.1	Percent of the content completed (0–100)
completed	BOOLEAN	True	Whether the session passed the completion threshold

subscribers.csv — Subscriber Profiles

Column	Type	Example	Description
subscriber_id	VARCHAR	SUB100001	Unique subscriber ID
plan_type	VARCHAR	Standard	Basic with Ads / Standard / Premium
monthly_price_usd	DECIMAL	15.49	Monthly subscription price in USD
country / region	VARCHAR	Italy / Europe	Geographic location
age	INT	35	Subscriber age in years
tenure_months	INT	50	Months since signup (pre-calculated)
is_active	BOOLEAN	True	Whether the account is currently active
churn_date	DATE	2022-12-08	Date of churn (blank if still active)

titles.csv — Content Catalogue

Column	Type	Example	Description
title_id	VARCHAR	TTL200001	Unique title ID
type	VARCHAR	TV Show	Movie or TV Show
primary_genre	VARCHAR	Fantasy	Main genre of the title
language	VARCHAR	Korean	Primary language
is_original	BOOLEAN	True	Whether it is a StreamFlix Original
license_cost_usd	DECIMAL	3,074,516	Cost to license or produce the title
total_watch_hours	DECIMAL	1,767.2	Total hours watched across all subscribers
total_plays	INT	52	Total viewing sessions started

3. Project Tasks & Deliverables

The project is divided into 4 phases. Complete them in order — each phase builds on the last. The estimated time per phase is shown below.

Phase	Deliverable	Skills Used	Difficulty
Phase 1 (Week 1)	Data Loading, Cleaning & Quality Report	Python/Excel, Pandas	Beginner–Intermediate
Phase 2 (Week 2)	Exploratory Data Analysis (EDA) with Charts	Pandas, Matplotlib / Seaborn	Intermediate
Phase 3 (Week 2–3)	Business KPI Calculations & Insights	Python / SQL / Excel	Intermediate
Phase 4 (Week 3–4)	Dashboard + Management Summary Report	Power BI / Excel / Python	Intermediate

3.1 Phase 1 — Data Cleaning & Quality Check (Est. 5–6 hours)

Before any analysis, you must understand and clean the data. A real analyst never trusts raw data blindly.

What to Do in Phase 1

1. Load all 6 CSV files into Python (pandas) or Excel.
2. For each table, run a basic profile: row count, column count, and data types.
3. Check for missing values in every column — report which columns have nulls and what % of rows are affected.
4. Check for duplicate rows — especially duplicate watch_id in watch_history.csv and subscriber_id in subscribers.csv.
5. Validate data types — watch_date and signup_date should be dates, not strings; durations and prices should be numeric.
6. Check for outliers in watch_duration_min — are there sessions longer than the content itself? Flag them.

7. Verify referential integrity — do all subscriber_id and title_id values in watch_history.csv exist in their parent tables?
8. Write a 1-page Data Quality Report summarising your findings.

Specific checks to run:

- subscribers.csv: For churned subscribers, confirm churn_date is after signup_date. Flag any row where it is not.
- subscribers.csv: Confirm active subscribers (is_active = True) have a blank churn_date. Flag mismatches.
- watch_history.csv: Confirm completion_pct roughly equals watch_duration_min ÷ content_duration_min × 100.
- reviews.csv: Confirm sentiment only contains Positive / Neutral / Negative. Flag anything else.
- watchlist.csv: Does every watchlist entry link to a real subscriber and a real title?
- subscribers.csv: Are there subscribers with very short tenure but marked active? That is fine — just note it.

Deliverable: A Python notebook (or Excel sheet) with your cleaning code and a written summary of findings. Name it: `Phase1_DataCleaning_[YourName].ipynb`

3.2 Phase 2 — Exploratory Data Analysis (Est. 6–8 hours)

EDA is about understanding patterns, distributions, and relationships in the data. Use charts to tell the story. Create the following charts and write 2–3 sentences of observation for each:

- Monthly viewing volume — a bar or line chart showing number of sessions per month across the timeline. Highlight the busiest months.
- Monthly watch hours trend — same as above but using watch_duration_min (converted to hours). Compare year-over-year.
- Genre-wise watch-hours share — a pie or horizontal bar chart. Which genre earns the most watch time?
- Content type split — a donut chart showing the share of hours for Movies vs. TV Shows.
- Top 10 countries by watch hours — a horizontal bar chart.
- Subscriber plan distribution — a pie chart. How many Basic vs. Standard vs. Premium subscribers?
- Device usage — a bar chart showing which devices people watch on most.
- Age distribution of subscribers — a histogram. Which age group watches most?
- Completion rate by genre — a bar chart. Which genres get finished?
- Review sentiment breakdown — a bar chart of Positive / Neutral / Negative review counts.

Deliverable: A Python notebook with all 10 charts. Save charts as PNG images too. Name it: `Phase2_EDA_[YourName].ipynb`

3.3 Phase 3 — KPI Calculations (Est. 6–8 hours)

Now calculate the business metrics that leadership actually asks about. These are the numbers that go into board reports.

KPI / Metric	How to Calculate	Why It Matters
Total Watch Hours	$\text{SUM}(\text{watch_duration_min}) \div 60$	Total engagement / size of the platform
Active Rate	$\text{Active Subscribers} \div \text{Total} \times 100$	Target > 70%. Health of the subscriber base
Churn Rate	$\text{Inactive Subscribers} \div \text{Total} \times 100$	Target < 30%. Higher = lost revenue
Avg Completion Rate	$\text{AVG}(\text{completion_pct})$	Target > 60%. Are people finishing content?
Monthly Recurring Revenue	$\text{SUM}(\text{monthly_price})$ for active subs	Actual recurring money earned
ARPU	$\text{MRR} \div \text{Active Subscribers}$	Average revenue per active user
Avg Watch Time / Subscriber	$\text{Total Watch Hours} \div \text{Active Subs}$	How engaged the average subscriber is
Watchlist Conversion	$\text{Watched} \div \text{Watchlist Entries} \times 100$	Target > 40%. Intent turning into viewing
Hit Concentration	$\text{Plays from top 10\% titles} \div \text{Total} \times 100$	Concentration risk if a few titles carry all plays
Originals Share of Hours	$\text{Hours on Originals} \div \text{Total Hours} \times 100$	How much engagement Originals drive

Intermediate Challenge: Cohort Retention (Optional, Bonus)

- Group subscribers by their signup month (cohort). For each cohort, measure how many were still active 3, 6 and 12 months later. This reveals retention — the single most important metric for any subscription business. **Hint:** combine `signup_date` with `watch_history` activity dates (or `churn_date`) using a `pandas pivot_table` or `groupby`.

Deliverable: A Python notebook (or Excel workbook) with all KPI calculations clearly labelled. Name it: `Phase3_KPIs_[YourName].ipynb`

3.4 Phase 4 — Dashboard & Final Report (Est. 8–10 hours)

The final phase brings everything together into a dashboard and a written report for leadership.

Dashboard Requirements (Power BI or Excel)

Build a dashboard with the following 5 pages / sections:

- Page 1 — Engagement Overview: monthly watch-hours trend, total subscribers, completion-rate card, churn-rate card.
- Page 2 — Content Performance: watch hours by genre, top 10 titles by hours, completion rate by genre.
- Page 3 — Subscriber Insights: plan/segment breakdown, top 10 countries map, new subscribers per month.
- Page 4 — Experience: device breakdown, rating distribution, review sentiment analysis.

- Page 5 — Catalogue & Investment: Originals vs. Licensed split, watch hours per \$1K spend by genre, upcoming licence expiries.

Each page must have: a title, at least 2 charts, at least 1 filter/slicer (e.g. date range, genre, or plan), and a text box with 1–2 key insights.

Management Summary Report

Write a 2–3 page report (in Word or PDF) addressed to the CEO of StreamFlix. The report should include:

- Executive Summary (5–6 sentences): What is the overall health of the platform?
- Top 3 Findings: What are the most important things you discovered?
- Risks Identified: Which numbers are concerning (e.g. churn, low completion, over-spend)?
- Opportunities: Where should the business focus to grow watch time and retention?
- Recommendations: At least 3 specific, actionable suggestions.

Important: Write in plain language. Avoid jargon. A manager who does not know Python should be able to read and understand your report.

Deliverable: Dashboard file (.pbix or .xlsx) + Management Report (.docx or .pdf). Name them: [Phase4_Dashboard_\[YourName\]](#) and [Phase4_Report_\[YourName\]](#).

4. Evaluation Rubric

Your submission will be evaluated on the following criteria. Total marks: 100.

Criteria	Marks	What We Look For
Phase 1: Data Cleaning & QC Report	15	Correct null/duplicate checks, clear documentation, logical conclusions.
Phase 2: EDA Charts (10 charts)	20	Charts clearly labelled, correct chart type, meaningful observations.
Phase 3: KPI Calculations	25	All 10 KPIs calculated correctly; formulas right; values make business sense.
Phase 4: Dashboard Quality	20	Dashboard is clean, filters work, data is accurate, all 5 pages complete.
Phase 4: Management Report	15	Clear writing, correct numbers cited, actionable recommendations, no jargon.
Bonus: Cohort Retention	+5	Correctly grouped by signup month; retention pattern is visible.

5. Submission Checklist

Before submitting, verify every item below is complete:

- [Phase1_DataCleaning_\[YourName\].ipynb](#) — Notebook with cleaning code and QC report
- [Phase2_EDA_\[YourName\].ipynb](#) — Notebook with all 10 charts (charts also saved as PNG)
- [Phase3_KPIs_\[YourName\].ipynb](#) — Notebook or Excel with all 10 KPI calculations
- [Phase4_Dashboard_\[YourName\].pbix/.xlsx](#) — Dashboard with 5 pages and slicers

- [Phase4_Report_\[YourName\].docx/.pdf](#) — 2–3 page management summary report
- All files named correctly with your name
- No hardcoded numbers in your report — every figure should trace back to your analysis
- Your report does NOT share any raw data files — only your analysis outputs

6. Tools & Resources

You may use any of the following tools. Python is recommended for data cleaning and KPI calculations.

Recommended Tool Stack

- **Python (pandas, matplotlib, seaborn)** — data cleaning, EDA, KPI calculations.
- **Jupyter Notebook** — for running and documenting your Python code.
- **Power BI Desktop (free)** — for the final dashboard.
- **Excel (optional)** — acceptable alternative to Power BI for the dashboard.
- **SQL (optional)** — the `schema_and_sql.sql` file has 12 ready-to-run queries you can study and modify.
- **VS Code or PyCharm** — recommended Python editors.

Helpful Python libraries to install:

`pip install pandas matplotlib seaborn openpyxl numpy`

Hints for Common Challenges

- **Date parsing:** `pd.to_datetime(df['watch_date'])` to convert date strings.
- **Joining tables:** `pd.merge(watch_history, titles, on='title_id', how='left')` to link sessions to content.
- **Grouping by month:** `df['watch_date'].dt.to_period('M').value_counts().sort_index()`.
- **Watch hours:** `df['watch_duration_min'].sum() / 60`.
- **Churn rate:** `(~df['is_active']).mean() * 100`.
- **Currency formatting:** `f'${value:,.0f}'` for revenue display in Python.

7. Important Notes

- **Academic integrity:** All analysis must be your own. You may discuss approaches with peers, but code and reports must be written independently.
- **If you get stuck,** describe what you are trying to do in your notebook as a comment — partial work that shows your thinking still earns partial marks.
- **You do NOT need to use all 6 tables for every phase.** Focus on the most relevant tables for each task.
- **The `schema_and_sql.sql` file** contains 12 queries that perform exactly the kind of analysis required in Phase 3. Study them even if you are using Python.

Good luck, and happy analysing!

This is exactly the kind of work junior and mid-level data analysts do every day at streaming companies. The skills you build here — data cleaning, KPI calculation, visualisation, and business communication — are directly applicable in the industry.